



Public-sector pension plans and the discount rate assumption: The role of political incentives[☆]

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ABSTRACT

We provide evidence that political pressure creates incentives for defined benefit pension plans to use higher discount rates, thus artificially reducing the reported cost of these benefit promises. We generate our inferences using a unique panel dataset for all local pension plans from Pennsylvania for 2003–2013, and by comparing the differential response of plans managed by appointed versus elected officials during a period where there was downward pressure on expected investment returns. The differential response we document is 29 basis points, which equates to a reduction in the reported liability of approximately 4.1% due to political incentives.

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1. Introduction

State and local governments carry substantial amounts of off balance-sheet debt in the form of defined benefit (DB) pension promises, with estimates of unfunded liabilities exceeding \$4 trillion at the state level (Novy-Marx and Rauh, 2011; Rauh, 2019). An ongoing controversy with DB plans is the use of a discount rate assumption that does not reflect the essentially guaranteed nature of these promises, but rather is based on the expected investment performance of the assets held in the trust. Within this literature, a number of studies examine the use of discretion in setting assumptions, as the manipulation of the discount rate is a powerful lever through which pension costs can be obscured (e.g. Naughton et al., 2015; Novy-Marx, 2013).

We contribute to this literature by examining the role of political pressure in influencing the choice of a discount rate. Elected

officials, regardless of party, have incentives to promise generous retirement benefits to public-sector workers to curry favor, while also having incentives to obscure the true magnitude of these promises from taxpayers and other stakeholders. Higher discount rates lower costs by reducing the present value of expected future pension payments (Naughton, 2019). Because differences between actuarial assumptions and actual performance are amortized over long periods (e.g., 30 years), the financial effect of choosing too high of a discount rate is essentially deferred for several years, at which point the elected politicians who made these decisions could be out of office. Thus, the incentive to avoid politically fraught decisions such as cutting benefits or raising taxes, coupled with a lack of voter awareness about actuarial assumptions, can result in officials subject to political pressure selecting higher discount rates.

To test this hypothesis, we use a high-quality administrative dataset that covers the universe of local pension plans in Pennsylvania for the period 2003 through 2013. The advantage of using this dataset of local plans is that we can compare plans where the level of political pressure is different, but where the plans are subject to the same actuarial and accounting requirements. There was pressure on public pension plans over the 2003–2013 period to lower their discount rate due to the fact that gross projected investment returns (which are the basis for the discount rate

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assumption) were lower than in earlier periods.¹ As previously noted, lowering the discount rate can lead to increased costs, and is therefore something that could be moderated by political pressure. In our analyses, we contrast the effects of political incentives on municipal plans managed by elected officials (“treatment”) with municipal authority plans managed by appointed officials (“control”). Across a variety of specifications, we find that pension plans run by appointed officials decrease their discount rate by about 29 basis points more than those run by elected officials. A discount rate difference of 29 basis points equates to a reduction in the pension liability of approximately 4.1% assuming an average duration of 15 years, which is typical for public sector pension plans (see e.g. Naughton et al., 2015). Additionally, we confirm the results of prior work that underfunded plans select higher discount rates that obscure the extent of underfunding (e.g. Giertz and Papke, 2007). These results suggest that political pressure encourages the use of more aggressive discount rate assumptions, which has the effect of reducing the current cost of pension promises.²

2. Institutional background

We exploit three unique features of local pension systems in the state of Pennsylvania to generate our inferences. First, Pennsylvania has ~1,400 retirement systems accounting for 40 percent of the nation’s public-sector retirement systems, three times more than any other state.³ The existence of such a large number of retirement systems can be attributed to Pennsylvania’s large number of local governments, the second highest in the country. Second, unlike other states, most local governments in Pennsylvania establish separate plans for their employees and are not part of a larger state-wide plan.⁴

Third, there are over 1,500 active municipal authorities in the state (DCED, 2015) and several have DB plans of their own. Municipal authorities (or special districts) are typically organized around the provision of a particular function, such as water or sewer or parking. Authority board members are appointed by the governing body of the municipalities where they provide service for five-year overlapping terms. Therefore, unlike a municipality’s governing body that can change abruptly following an election, appointed members serving on an authority board are insulated from the electoral will of voters. However, even though there are significant differences in the influence of political pressure on municipal governments vis-à-vis municipal authorities, there are no differences in the financial reporting requirements across plans or in the geographic coverage of municipal authorities across the state.⁵

¹ See <https://fred.stlouisfed.org/series/FEDFUNDS>, <https://fred.stlouisfed.org/series/BOERUKQ>, and <https://fred.stlouisfed.org/series/IR3TIB01EZM156N> for some examples of interest rate cuts during the Great Recession and <https://www.invesco.com/pdf/HAR-BRO-1.pdf> for data on returns across asset classes between 2005 and 2014.

² Numerous papers, including Kelley (2014), Elder and Wagner (2015), and Bagchi (Forthcoming) examine the impact of political incentives on the funding of public pensions.

³ Source: <https://www.census.gov/prod/2013pubs/g11-aspp-sl.pdf> (Ref. Table 5a). Of the 3,418 public-employee retirement systems in the United States, 1,422 (or, 41.6%) are local systems from Pennsylvania.

⁴ Further details on Pennsylvania’s municipal pensions are available in Bagchi (2019, Forthcoming).

⁵ Further details on the accounting and actuarial requirements for municipal pensions are available in Dambra et al. (2021).

3. Empirical analysis

3.1. Construction of variables and data sources

Data regarding local pension plans from Pennsylvania are available biennially from 2003 through 2013 from the Public Employee Retirement Commission (PERC). The data include the name of the entity offering the plan, the employee group covered, coverage under Social Security, the number of active members and beneficiaries, the funded ratio, and the interest rate assumed in discounting actuarial liabilities. Summary statistics are presented in Table A.1.⁶

3.2. Empirical specification

We use a differences-in-differences (DiD) approach to examine whether political incentives induce elected officials in charge of municipal pension plans to assume higher discount rates than plans run by municipal authorities and exploit the change in the strength of these incentives stemming from an economy-wide decline in interest rates. We use 2003, the starting year of our sample, as the “pre-period” and 2013, the final year of our sample, as the “post-period”. Thus,

$$Y_{imt} = \alpha + \beta_1 * M_i + \beta_2 * Post_t + \beta_3 * M_i * Post_t + \beta_4 * C_{it} + \beta_5 * X_{it} + \beta_6 * Z_{mt} + \lambda_c + \varepsilon_{imt} \quad (1)$$

where: M_i is a dummy variable set to 1 if plan i is operated by a municipality m and zero otherwise; $Post_t$ is a dummy variable set to 1 for the year 2013 and zero for the year 2003; C_{it} are a set of dummy variables indicating which group of employees are covered by the plan (e.g., policemen or firefighters or nonuniformed personnel); X_{it} includes plan-specific controls for coverage under Social Security, presence of collective bargaining, the funded ratio, and the number of active members and beneficiaries; Z_{mt} are time-variant controls at the municipal level for the unemployment rate, debt per capita, taxes as a share of per capita income, and the log of per capita income⁷; λ_c are county fixed effects; and ε_{imt} is the error term. We cluster standard errors at the county level all throughout.

4. Results

4.1. Baseline specification

Before presenting the regression results, a cross-tabulation of the discount rate used by municipalities and municipal authorities in 2003 and 2013 is insightful. Over this period, the average discount rate assumed by authorities fell by 64 basis points (from 6.79 to 6.15 percent), whereas that assumed by municipalities fell by 35 basis points (from 7.04 to 6.69 percent), yielding a differential of 29 basis points in a DiD framework. Although this tabulation merely provides a stab at the raw data, a multivariate regression including numerous controls matches these findings.

We present our baseline estimates in Table 1. Column (1) is parsimonious and only includes a dummy variable for municipal authorities, a dummy variable for the “post-period”, and the

⁶ The reported funded ratio in Table A.1 is based on discounting future liabilities at a rate averaging 7.0%. If we discounted liabilities using a rate of 3.5%, reflecting the essentially guaranteed nature of pensions, the median funded ratio declines from 94 percent to 67 percent.

⁷ The percentage of employees organized under collective bargaining is for 1982 from the Employment Summary Statistics of Census of Governments. Per capita income and unemployment rate are from the Census and the ACS. Taxes as a share of per capita income and debt per capita are based on data from 2003 to 2013 from the DCED.

Table 1

Baseline specification: DiD effects of stronger incentives for elected officials on the discount rate chosen by pension plans.

	(1)	(2)	(3)	(4)	(5)	(6)
Dummy variable for municipality	−0.00878 (0.145)	−0.00536 (0.126)	−0.00187 (0.111)	−0.160 (0.117)	−0.144 (0.109)	−0.124 (0.0996)
Dummy variable for "post-period"	−0.638*** (0.0410)	−0.666*** (0.0421)	−0.630*** (0.0421)	−0.778*** (0.0475)	−0.789*** (0.0482)	−0.751*** (0.0476)
Dummy variable for municipality	0.291*** (0.0391)	0.281*** (0.0387)	0.262*** (0.0382)	0.299*** (0.0426)	0.292*** (0.0419)	0.274*** (0.0408)
X dummy variable for "post-period"		−4.316*** (0.385)			−2.814*** (0.371)	
Funded ratio as reported by plans themselves			−8.493*** (0.611)			−6.625*** (0.542)
Funded ratio adjusted assuming a discount rate of 3.5% and a maturity of liabilities = 11 years				−0.119*** (0.0422)	−0.0836** (0.0413)	−0.0514 (0.0388)
Coverage of employees under Social security (0 = No, 1 = Yes)				0.189** (0.0775)	0.190** (0.0739)	0.182*** (0.0685)
Fraction of employees covered under collective bargaining				0.185*** (0.0282)	0.150*** (0.0256)	0.116*** (0.0238)
Log of number of active members				0.0689*** (0.0206)	0.0670*** (0.0204)	0.0621*** (0.0204)
Log of number of beneficiaries				0.00199 (0.00473)	0.00257 (0.00452)	0.00298 (0.00427)
Unemployment rate				0.0183 (0.0302)	0.0232 (0.0292)	0.0261 (0.0273)
Debt per capita				17.78*** (4.428)	16.34*** (4.221)	14.44*** (3.910)
Ratio of taxes to per capita income				0.333*** (0.0925)	0.309*** (0.0859)	0.273*** (0.0782)
Log of per capita income				3.384*** (0.918)	3.963*** (0.852)	4.560*** (0.766)
Constant	7.346*** (0.126)	7.776*** (0.104)	7.918*** (0.103)			
Number of observations	3951	3951	3951	3951	3951	3951
R ²	0.30	0.35	0.41	0.41	0.42	0.47

Standard errors clustered by county in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable and all control variables have been winsorized at the 2.5% and 97.5% levels.

Table 2

Baseline specification, augmented with a control for the change in level of assets between 2007 and 2009 as a proxy for the asset allocation.

	(1)	(2)	(3)	(4)	(5)	(6)
Dummy variable for municipality	−0.0565 (0.145)	−0.0539 (0.124)	−0.0479 (0.111)	−0.163 (0.116)	−0.146 (0.107)	−0.128 (0.0977)
Dummy variable for "post-period"	−0.636*** (0.0411)	−0.666*** (0.0425)	−0.629*** (0.0421)	−0.732*** (0.0492)	−0.743*** (0.0504)	−0.706*** (0.0495)
Dummy variable for municipality	0.291*** (0.0392)	0.281*** (0.0388)	0.263*** (0.0381)	0.293*** (0.0430)	0.285*** (0.0423)	0.268*** (0.0410)
X dummy variable for "post-period"		−4.465*** (0.331)			−3.078*** (0.318)	
Funded ratio as reported by plans themselves			−8.254*** (0.542)			−6.558*** (0.471)
Funded ratio adjusted assuming a discount rate of 3.5% and a maturity of liabilities = 11 years			−0.0157*** (0.000900)	−0.0150*** (0.00119)	−0.0154*** (0.00112)	−0.0148*** (0.00102)
Change in the actuarial assets between 2007 and 2009	−0.0163*** (0.00110)	−0.0167*** (0.000998)		−0.143*** (0.0427)	−0.105** (0.0410)	−0.0757* (0.0386)
Coverage of employees under Social security (0 = No, 1 = Yes)				0.130* (0.0757)	0.130* (0.0721)	0.124* (0.0672)
Fraction of employees covered under collective bargaining				0.228*** (0.0243)	0.191*** (0.0223)	0.159*** (0.0213)
Log of number of active members				0.000498 (0.0206)	−0.00352 (0.0206)	−0.00558 (0.0205)
Log of number of beneficiaries				0.000384 (0.00468)	0.000970 (0.00449)	0.00137 (0.00428)
Unemployment rate				0.0237 (0.0271)	0.0291 (0.0259)	0.0313 (0.0242)
Debt per capita				14.91*** (4.109)	13.26*** (3.864)	11.64*** (3.588)
Ratio of taxes to per capita income				0.328*** (0.0889)	0.301*** (0.0811)	0.268*** (0.0739)
Log of per capita income				3.589*** (0.893)	4.228*** (0.809)	4.751*** (0.732)
Constant	7.437*** (0.134)	7.883*** (0.110)	7.989*** (0.107)			
Number of observations	3951	3951	3951	3951	3951	3951
R ²	0.38	0.44	0.49	0.47	0.50	0.54

Standard errors clustered by county in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable and all control variables have been winsorized at the 2.5% and 97.5% levels.

interaction of the two, along with dummy variables for municipal class, the employee-groups covered, and county fixed effects.

Column (2) adds a control for the funded ratio as reported by the plans themselves. Given that the reported ratio depends on

the discount rate chosen – our dependent variable – we “unskew” the reported ratio by using a common discount rate of 3.5% across all plans. Cols. (4)–(6) replicate the patterns in cols. (1)–(3) but we now add an extensive battery of municipal- and plan-specific controls.

As the results in the second and third rows of Table 1 indicate, although discount rates declined by about 70 basis points for municipal authority plans, the declines were smaller for municipal plans by about 29 basis points. The funded ratio of the plans also plays a role in the choice of discount rates with a one percent decline in the funded ratio associated with an increase in the discount rate chosen by about 3–4 basis points (when using numbers reported by plans themselves) and about 7–8 basis points (when using the “unskewed” numbers).⁸

4.2. The role of asset allocation

Since the discount rate is derived from expectations about future investment returns, it is possible that changes in this assumption are because of underlying differences in the composition of assets held in the pension trust. As the PERC dataset does not have detailed information on asset allocations, we construct a proxy of the asset mix by exploiting the fact that bond prices went up during 2008 whereas equity prices took a large hit.^{9, 10} The decline in equity markets during 2008 and early 2009 resulted in the nation's pension plans suffering a drop of about 20 percent in their assets' value on average during fiscal year 2009.¹¹ However, actual investment losses differed significantly from those averages in part because the asset mix varied across plans.¹² Therefore we can use the drop in the value of assets between 2007 and 2009 as a proxy for the asset allocation of the plan¹³ and in Table 2, we augment our baseline specification with the inclusion of a change in the level of assets between 2007 and 2009.

The results in Table 2 confirm the robustness of our earlier finding. Even though we obtain a negative and statistically significant coefficient on the change in assets between 2007 and 2009, we continue to find that municipal plans use discount rates higher by about 29 basis points.

5. Conclusions

Our study supplements the results of prior work by documenting that local (i.e., not state level) underfunded pension plans also choose higher discount rates. Moreover, our results are starker for poorly funded pension plans with municipal plans that were funded less (more) than the median in 2003 choosing a discount rate that is about 41 (16) basis points higher than authority plans by 2013.

⁸ The dependent variable and control variables have been winsorized at the 2.5% and 97.5% levels to reduce the influence of outliers. However, our results are not sensitive to either the decision to winsorize or the particular level of winsorization chosen, with these additional results provided in Appendix Tables A.2 and A.3.

⁹ http://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/histretSP.html

¹⁰ Even though we create this proxy using data for 2008, investment compositions for public plans are relatively consistent over time (e.g., Naughton et al., 2015).

¹¹ https://www.pewtrusts.org/~/media/legacy/uploadedfiles/pes_assets/2011/pewpensionsretireebenefitspdf.pdf

¹² <https://www.brookings.edu/research/the-financial-crisis-effects-on-the-alternatives-for-public-pensions/>

¹³ Additional analysis in the Appendix confirms that the change in asset values and funding ratios between 2007 and 2009 were highly atypical of all other two-year periods between 2003 and 2013.

The key contribution of this paper, however, lies in demonstrating the role that political incentives play in influencing the discount rate assumption. Over the 2003 – 2013 period, we find that appointed officials in charge of municipal authority plans were more likely to reduce their assumed discount rate compared with elected officials in charge of municipal plans. This result is robust to numerous checks, including clustering standard errors at the municipal level, introducing municipality fixed effects, eliminating all spatial fixed effects, or controlling for the change in funding ratio between 2007 and 2009 (Ref. Table A.4). The overall message is straight-forward: political incentives matter when it comes to the choice of a discount rate.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.econlet.2021.109914>.

References

- Bagchi, Sutirtha, 2019. The effects of political competition on the generosity of public-sector pension plans. *J. Econ. Behav. Organ.* 164 (439).
- Bagchi, Sutirtha, The Effects of Political Competition on the Funding of Public-Sector Pension Plans. *Financ. Manag.* <http://dx.doi.org/10.1111/fima.12335>. Forthcoming.
- Dambra, Michael, Even-Tov, Omri, Naughton, James P., 2021. The Economic Consequences of GASB Financial Statement Disclosure. Working Paper. University of Virginia, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3538354.
- Department of Community and Economic Development (DCED), 2015. *Municipal Authorities in Pennsylvania*. Governor's Center for Local Government Services, Harrisburg, PA.
- Elder, Erick M., Wagner, Gary A., 2015. Political effects on pension underfunding. *Econ. Politics* 27 (1), 1–27.
- Giertz, J. Fred., Papke, E., 2007. Public pension plans: Myths and realities for state budgets. *Natl. Tax J.* 60 (2), 305–323.
- Kelley, Dashle G., 2014. The political economy of unfunded public pension liabilities. *Public Choice* 158 (1/2), 21–38.
- Naughton, James P., 2019. Regulatory oversight and trade-offs in earnings management: Evidence from pension accounting. *Rev. Account. Stud.* 24 (2), 456–490.
- Naughton, James P., Petacchi, Reining, Weber, Joseph P., 2015. Public pension accounting rules and economic outcomes. *J. Account. Econ.* 59 (2–3), 221–241.
- Novy-Marx, Robert., 2013. Logical implications of GASB's methodology for valuing pension liabilities. *Financ. Anal. J.* 69 (1), 26–32.
- Novy-Marx, Robert, Rauh, Joshua D., 2011. Public pension promises: How big are they and what are they worth?. *J. Financ.* 66 (4), 1211–1249.
- Rauh, Joshua D., 2019. Hidden Debt, Hidden Deficits: The 2019 Update. <https://www.hoover.org/news/hidden-debthidden-deficits-2019-update>.

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Table A.1: Summary Statistics

Variable	Units	Mean	Median	Standard deviation	Minimum	Maximum
Pension plan characteristics						
For municipal plans:						
Interest rate in 2003 (Unwinsorized)	In percent terms	7.04	7.00	0.682	5.00	9.00
Interest rate in 2003 (Winsorized as noted below)	In percent terms	7.04	7.00	0.660	5.50	8.00
Interest rate in 2013 (Unwinsorized)	In percent terms	6.66	7.00	1.019	3.00	8.50
Interest rate in 2013 (Winsorized as noted below)	In percent terms	6.69	7.00	0.958	5.50	8.00
Actuarial funded ratio	In percent terms	103.72	93.49	48.42	45.07	304.57
Number of active members		33.8	7	501.99	0	20540
Number of beneficiaries		33.2	3	593.10	0	23117
For plans run by municipal authorities:						
Interest rate in 2003 (Unwinsorized)	In percent terms	6.79	6.50	0.557	5.00	8.50
Interest rate in 2003 (Winsorized as noted below)	In percent terms	6.79	6.50	0.535	5.50	8.00
Interest rate in 2013 (Unwinsorized)	In percent terms	6.15	5.50	0.945	5.00	8.50
Interest rate in 2013 (Winsorized as noted below)	In percent terms	6.15	5.50	0.928	5.50	8.00
Actuarial funded ratio	In percent terms	104.73	98.43	36.60	45.07	304.57
Number of active members		21.2	9	37.95	0	361
Number of beneficiaries		9.16	2	24.38	0	212
Controls at the municipal level						
Unemployment rate	In percent terms	6.89	6.28	2.94	2.08	14.65
Debt per capita	In dollars	474.6	196.6	616.6	0	2135
Ratio of taxes to per capita income		1.20	1.08	0.55	0.36	2.60
Per capita income	In dollars	25510	23798	7816	14415	46643

Summary statistics for actuarial funded ratio, interest rate, Social security coverage, and number of active members and beneficiaries are based on biennial data from 2003–2013 from the PERC. Coverage under collective bargaining is based on the 1982 Employment Summary Statistics of Census of Governments. Unemployment rate and per capita income are from the 2007–11 and 2011–15 ACS. Debt per capita and ratio of taxes to per capita income are constructed based on annual data from the Pennsylvania Department of Community and Economic Development (DCED). Unless otherwise mentioned, all variables have been winsorized at the 2.5% and 97.5% levels.

Table A.2: Baseline specification: DiD effects of stronger incentives for elected officials on the discount rate chosen by pension plans

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Baseline specification – Winsorizing the dependent variable and all control variables at the 2.5% level						
Dummy variable for municipality	0.291***	0.281***	0.262***	0.299***	0.292***	0.274***
X Dummy variable for “Post-period”	(0.0391)	(0.0387)	(0.0382)	(0.0426)	(0.0419)	(0.0408)
Panel B: Without winsorizing the dependent variable and any of the control variables						
Dummy variable for municipality	0.291***	0.281***	0.262***	0.299***	0.292***	0.274***
X Dummy variable for “Post-period”	(0.0383)	(0.0399)	(0.0395)	(0.0384)	(0.0393)	(0.0389)
Panel C: Winsorizing the dependent variable and all of the control variables at the 1% level						
Dummy variable for municipality	0.268***	0.257***	0.244***	0.281***	0.273***	0.261***
X Dummy variable for “Post-period”	(0.0397)	(0.0397)	(0.0396)	(0.0431)	(0.0430)	(0.0427)
Panel D: Winsorizing the dependent variable and all of the control variables at the 5% level						
Dummy variable for municipality	0.291***	0.289***	0.261***	0.290***	0.288***	0.265***
X Dummy variable for “Post-period”	(0.0391)	(0.0396)	(0.0391)	(0.0434)	(0.0431)	(0.0415)

Standard errors clustered by county in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: All results include a dummy variable for municipal authorities, a dummy variable for the “Post” period, county fixed effects, dummies for municipal class and employee groups, e.g. non-uniformed personnel or policemen and firefighters. Col. (2) includes a control for the funded ratio as reported by plans; col. (3) unskews the reported ratio by using a common discount rate of 3.5% across all plans. Cols. (4)–(6) replicate the patterns in cols. (1)–(3) but additionally control for the coverage of employees under Social Security and presence of collective bargaining, the log of number of active members and beneficiaries, local unemployment rate, debt per capita, ratio of taxes to per capita income, and log of per capita income.

Table A.3: Baseline specification, augmented with a control for the change in level of assets between 2007 and 2009 as a proxy for the asset allocation

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Baseline specification – Winsorizing the dependent variable and all control variables at the 2.5% level						
Dummy variable for municipality	0.291*** (0.0392)	0.281*** (0.0388)	0.263*** (0.0381)	0.293*** (0.0430)	0.285*** (0.0423)	0.268*** (0.0410)
X Dummy variable for “Post-period”						
Panel B: Without winsorizing the dependent variable and any of the control variables						
Dummy variable for municipality	0.263*** (0.0400)	0.263*** (0.0405)	0.262*** (0.0406)	0.272*** (0.0443)	0.272*** (0.0446)	0.272*** (0.0446)
X Dummy variable for “Post-period”						
Panel C: Winsorizing the dependent variable and all of the control variables at the 1% level						
Dummy variable for municipality	0.268*** (0.0398)	0.257*** (0.0398)	0.245*** (0.0396)	0.274*** (0.0434)	0.266*** (0.0434)	0.255*** (0.0430)
X Dummy variable for “Post-period”						
Panel D: Winsorizing the dependent variable and all of the control variables at the 5% level						
Dummy variable for municipality	0.291*** (0.0392)	0.289*** (0.0397)	0.262*** (0.0387)	0.285*** (0.0439)	0.283*** (0.0435)	0.261*** (0.0417)
X Dummy variable for “Post-period”						

Standard errors clustered by county in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: All results include a dummy variable for municipal authorities, a dummy variable for the “Post” period, county fixed effects, dummies for municipal class and employee groups, e.g. non-uniformed personnel or policemen and firefighters, along with the change in level of assets between 2007 and 2009 as a proxy for the asset allocation. Col. (2) includes a control for the funded ratio as reported by plans; col. (3) unskews the reported ratio by using a common discount rate of 3.5% across all plans. Cols. (4)–(6) replicate the patterns in cols. (1)–(3) but additionally control for the coverage of employees under Social Security and presence of collective bargaining, the log of number of active members and beneficiaries, local unemployment rate, debt per capita, ratio of taxes to per capita income, and log of per capita income.

Table A.4: Robustness checks confirming the differential response of discount rate for municipalities relative to municipal authorities

	(1)	(2)	(3)	(4)	(5)	(6)
Baseline specification	0.291*** (0.0391)	0.281*** (0.0387)	0.262*** (0.0382)	0.299*** (0.0426)	0.292*** (0.0419)	0.274*** (0.0408)
RC1: Cluster standard errors at the municipality level	0.291*** (0.0383)	0.281*** (0.0399)	0.262*** (0.0395)	0.299*** (0.0384)	0.292*** (0.0393)	0.274*** (0.0389)
RC2: Introduction of municipality FE	0.298*** (0.0481)	0.293*** (0.0470)	0.280*** (0.0453)	0.305*** (0.0492)	0.300*** (0.0481)	0.286*** (0.0460)
RC3: Omit all FE, either county or municipality	0.288*** (0.0387)	0.277*** (0.0385)	0.257*** (0.0382)	0.296*** (0.0433)	0.288*** (0.0427)	0.270*** (0.0416)
RC4: Include a proxy of the plan's asset allocation, the change in funding ratio between 2007 & 2009	0.292*** (0.0391)	0.281*** (0.0389)	0.261*** (0.0382)	0.300*** (0.0426)	0.292*** (0.0419)	0.274*** (0.0407)
RC5: Splitting the sample of municipal plans based on their (recalibrated) funding ratio in 2003						
DiD estimate for municipal plans with funding less than the median in 2003	0.392*** (0.0374)	0.420*** (0.0367)	0.417*** (0.0367)	0.398*** (0.0409)	0.412*** (0.0399)	0.414*** (0.0388)
DiD estimate for municipal plans with funding more than the median in 2003	0.191*** (0.0469)	0.149*** (0.0480)	0.115** (0.0474)	0.202*** (0.0504)	0.178*** (0.0514)	0.142*** (0.0507)

Standard errors in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Unless otherwise specified, all results include a dummy variable for municipal authorities, a dummy variable for the “Post” period, county fixed effects, dummies for municipal class and employee groups. Col. (2) includes a control for the funded ratio as reported by plans; col. (3) unskews the reported ratio by using a common discount rate of 3.5% across all plans. Cols. (4)—(6) replicate the patterns in cols. (1)—(3) but additionally control for the coverage of employees under Social Security and presence of collective bargaining, the log of number of active members and beneficiaries, local unemployment rate, debt per capita, ratio of taxes to per capita income, and log of per capita income. All regressions (except in RC1) cluster standard errors at the county level, with RC1 clustering at the municipal level.

Table A.5: Change in level of assets between any two consecutive years in the data computed at various percentiles in the distribution

Current year	Prior year	N	p1	p5	p10	p25	p50	p75	p90	p95	p99
2005	2003	1666	-18.54	-4.12	0.99	8.52	16.16	29.76	51.29	80.63	224.48
2007	2005	2015	-13.16	-1.35	3.31	8.50	15.34	24.91	39.79	54.13	99.82
2009	2007	2015	-30.86	-21.95	-15.58	-6.55	5.29	15.56	25.73	34.81	61.39
2011	2009	2015	-17.67	-6.08	-0.91	6.13	12.89	22.74	36.55	48.55	71.25
2013	2011	2015	-22.26	-8.11	-2.15	4.74	10.88	17.61	26.20	31.72	45.26

This table supports the decision to use the change in level of assets as a proxy for the asset mix (particularly, the equity allocation) of the plan. We can also see that the distributions are different – in particular, there is a much longer left tail in the distribution of change in level of assets between 2007 and 2009 than what we see in other years. Recognize that also because of use of asset smoothing techniques, not all of the drop in the level of assets between 2007 and 2009 may be reflected in the level of assets reported in 2009.

Fig. A.1: Change in actuarial assets for each two-year period between 2003 and 2013 (e.g. 2003 and 2005, 2005 and 2007, etc.)

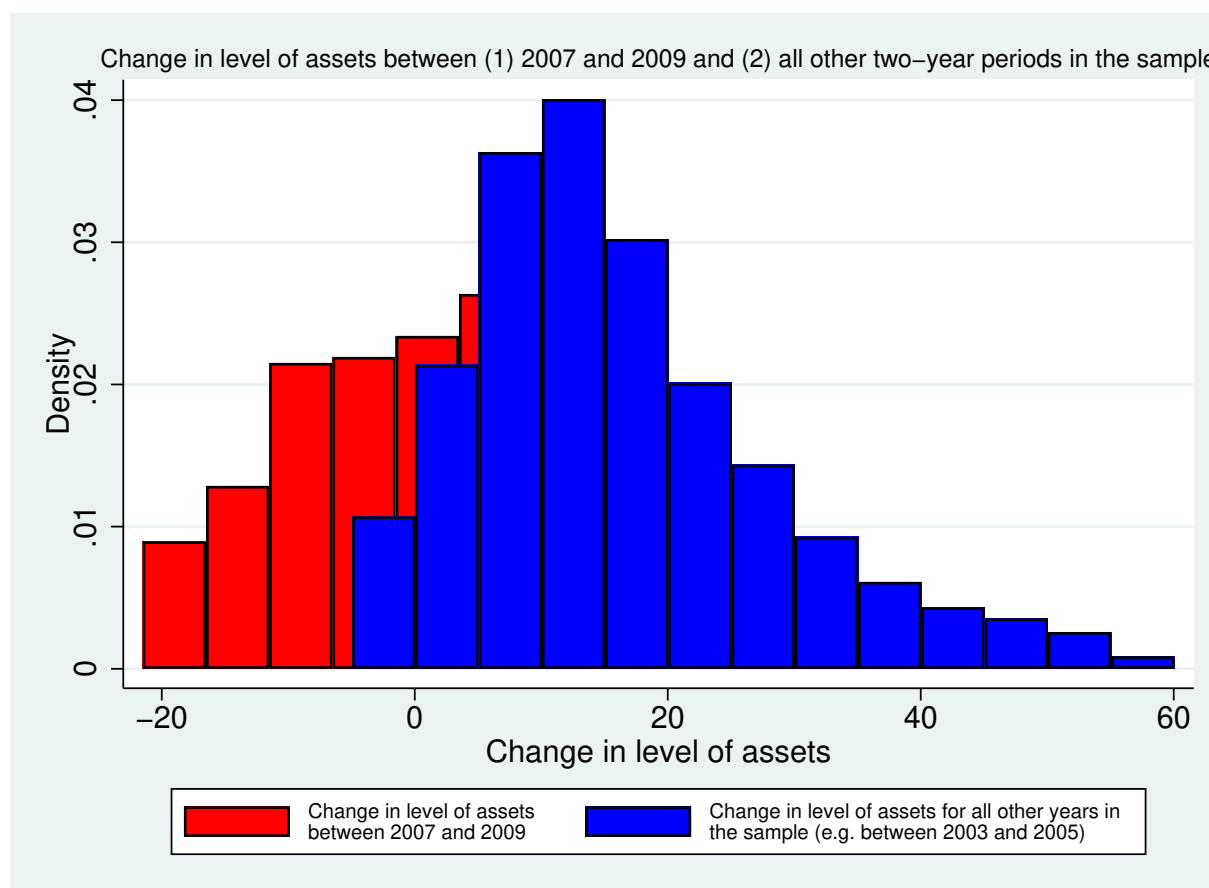


Table A.6: Change in funding ratio between any two consecutive years in the data computed at various percentiles in the distribution

Current year	Prior year	N	p1	p5	p10	p25	p50	p75	p90	p95	p99
2005	2003	1672	-69.20	-28.79	-17.65	-8.40	-0.60	8.19	19.99	30.37	69.25
2007	2005	2012	-43.40	-19.09	-11.18	-3.33	1.33	7.93	15.55	22.44	45.61
2009	2007	2012	-77.69	-41.20	-31.05	-17.72	-7.51	0.41	7.16	12.08	31.07
2011	2009	2013	-41.51	-17.85	-11.86	-5.45	0.00	6.04	14.70	21.04	45.34
2013	2011	2011	-37.99	-17.00	-11.94	-5.93	-1.13	3.20	8.86	14.47	39.29

This table supports the decision to use the change in funding ratio as a proxy for the asset mix (particularly, the equity allocation) of the plan. We can also see that the distributions are different – in particular, there is a much longer left tail in the distribution of change in funding ratios between 2007 and 2009 than what we see in other years.

Fig. A.2: Change in funding ratio for each two-year period between 2003 and 2013 (e.g. 2003 and 2005, 2005 and 2007, etc.)

